Final – Talk2ALMA

OpenACC
More Science, Less Programming

Talk2ALMA

Team Members





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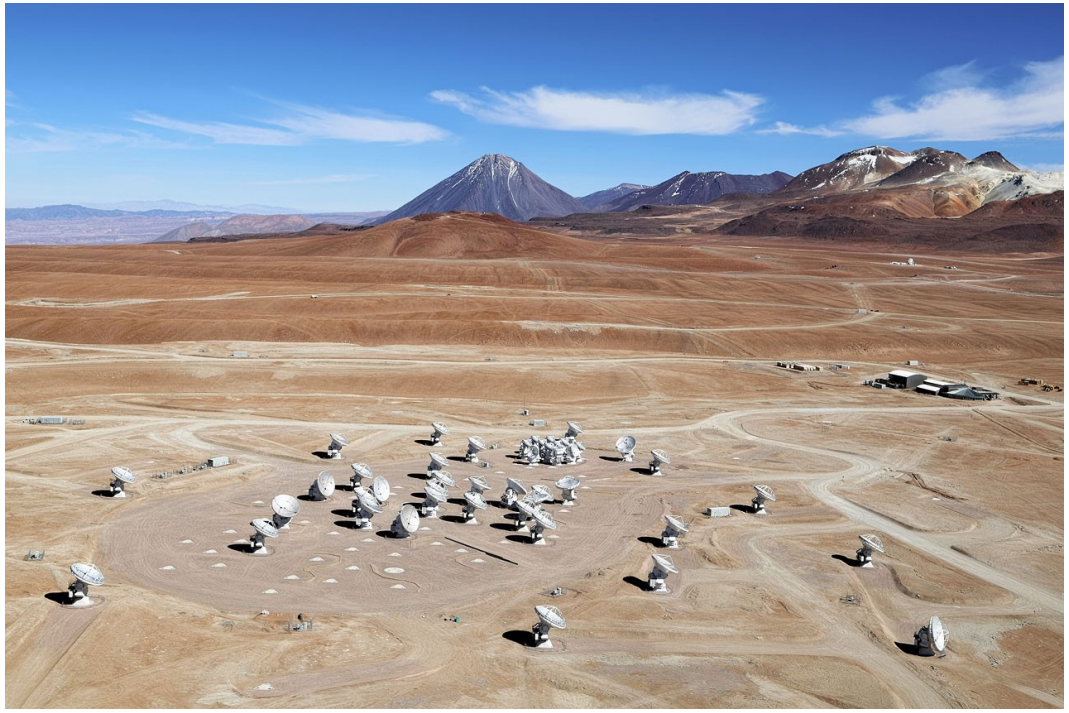
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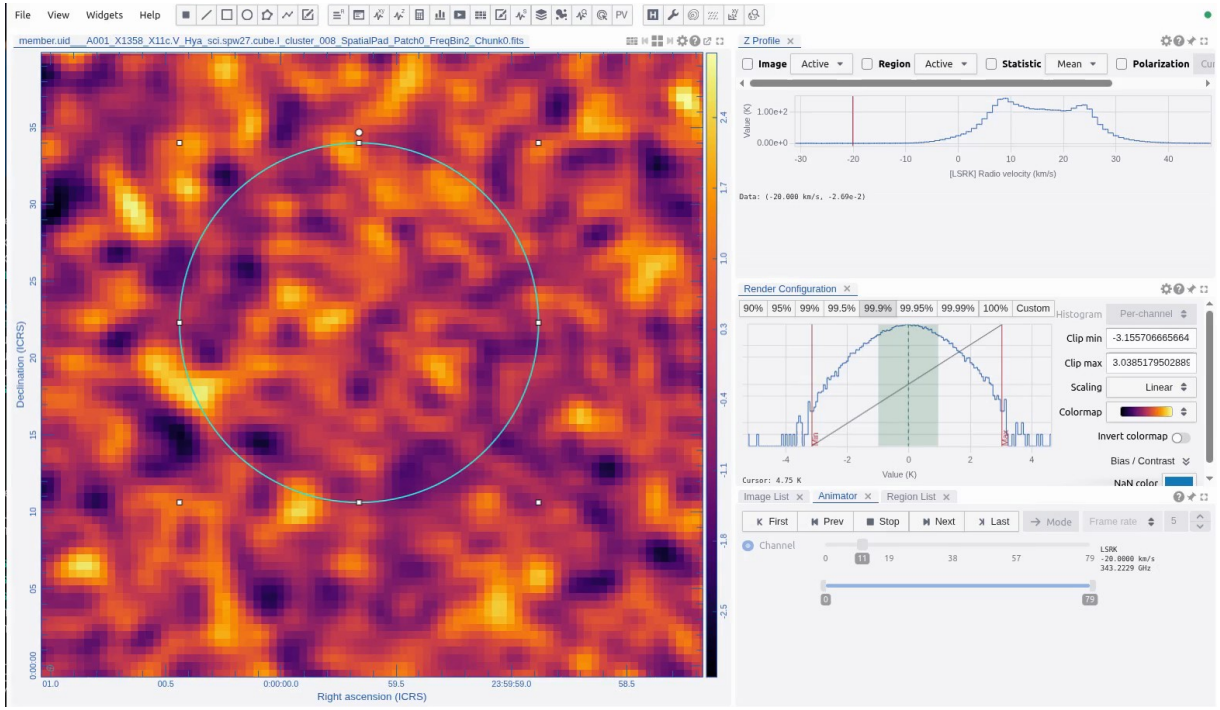
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How ALMA data are found today



ALMA, Chajnantor plateau, Chile (5,000 m): 66 antennas



data: a 2D image, or a cube that adds frequency and plays like a video

The archive is searched by

Project	Coordinates	Frequency
e.g. 2013.1.00546.S	RA, Dec or a source name	line, spw, resolution

Not by what the data show: "a rotating disk", "a bipolar outflow", "spiral arms". You must already know which project to open.

photo: ESO (eso1312a), CC BY 4.0

GPU training: Nano5 H100 vs g04 RTX 3080

	g04 (8 × RTX 3080)	Nano5 (8 × H100)
GPU	10 GB, PCIe	80 GB, NVLink all-to-all
Cube size	64×128×128 (1.05 M)	80×160×160 (2.05 M, 1.95×)
Max batch per GPU	32 (64 runs out of memory)	64
Precision / checkpoint	fp16, aug in fp32, all stages	bf16, aug in autocast, stage 0 only
Node cubes/s	49.5 (8 × 32)	102.2 (8 × 64)
Node, g04-equivalent	49.5	200
Per GPU, g04-equivalent	6.2	25.0

4.0×

faster per GPU than g04 in real training, at the same cube size

1.95×

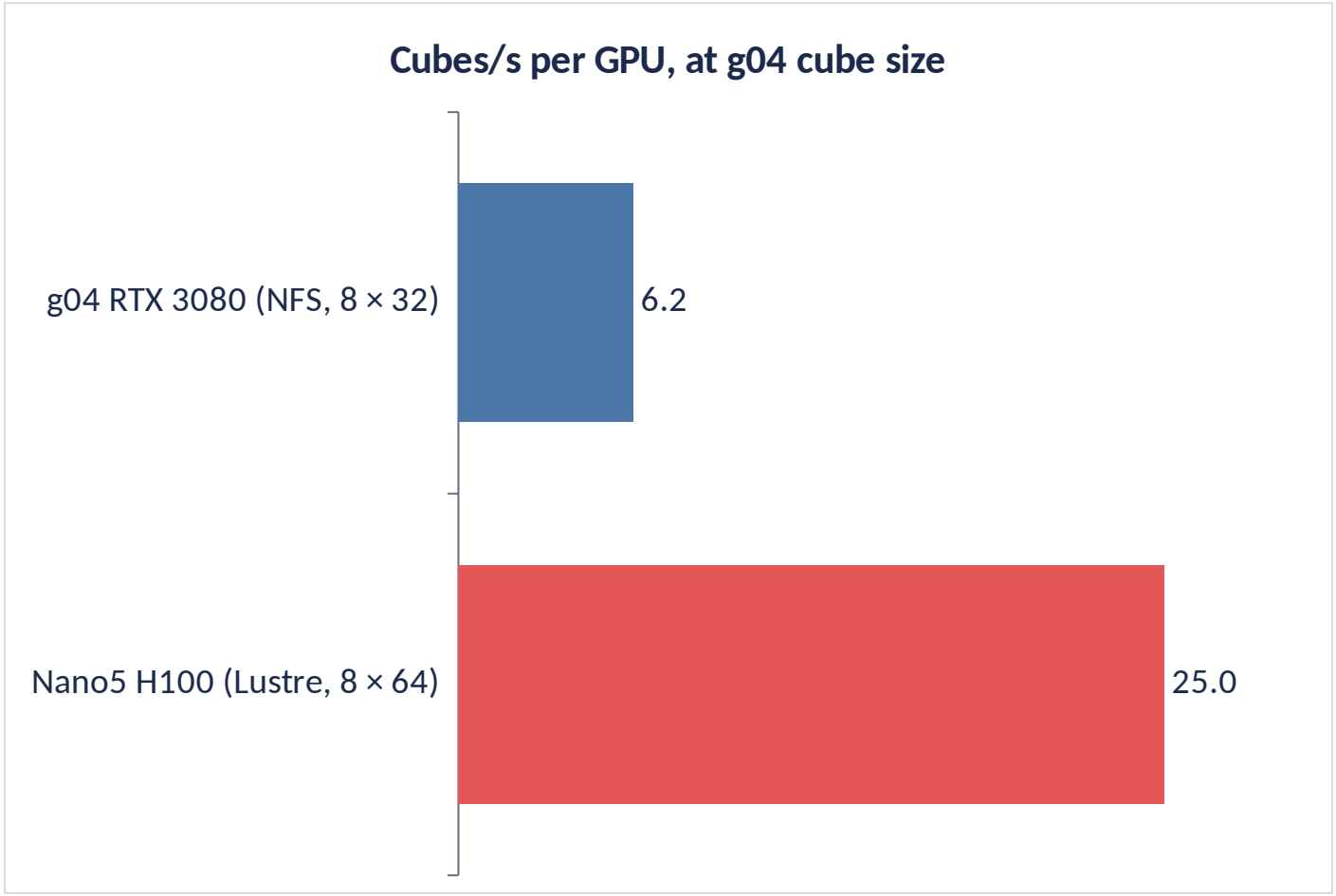
voxels per cube, and still 2.1× more cubes/s per node

2×

batch per GPU (64 vs 32), global batch 512; the 3080 runs out of memory at 64

g04-equivalent = cubes/s × 1.95 (64×128×128 cubes) · H100: job 414474, 8 GPUs, mean 5.01 s/step over steps 10→99 · same production code on both

Where the H100 gains come from



~4.4× memory bandwidth

H100 ~3.35 TB/s, 3080 ~760 GB/s; the 3D ConvNeXt is limited by memory bandwidth

bf16, augmentation in autocast too

on g04 the augmentation has to run in fp32 and takes 44% of each step

Gradient checkpoint on stage 0 only

80 GB is enough; the 3080 needs it everywhere, so backward recomputes the whole forward

NVLink all-to-all

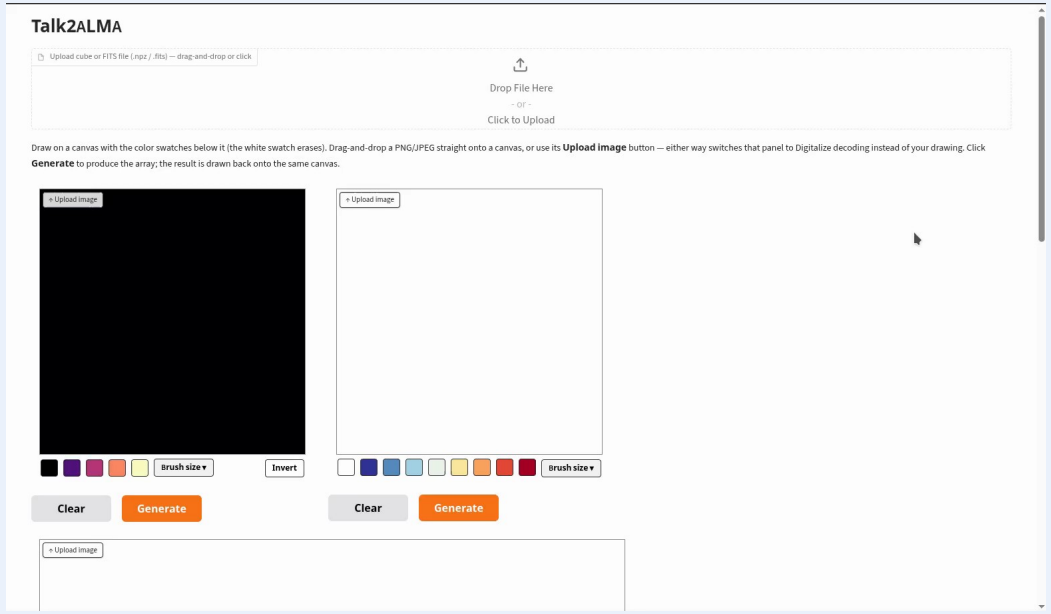
g04 is PCIe, but communication was only ~3% of a step there, not the main cause

per epoch: g04 42 min (488 steps), H100 18.7 min (224 steps); the datasets differ, so compare cubes/s across machines

Demo: drop a FITS cube, or draw a velocity field etc

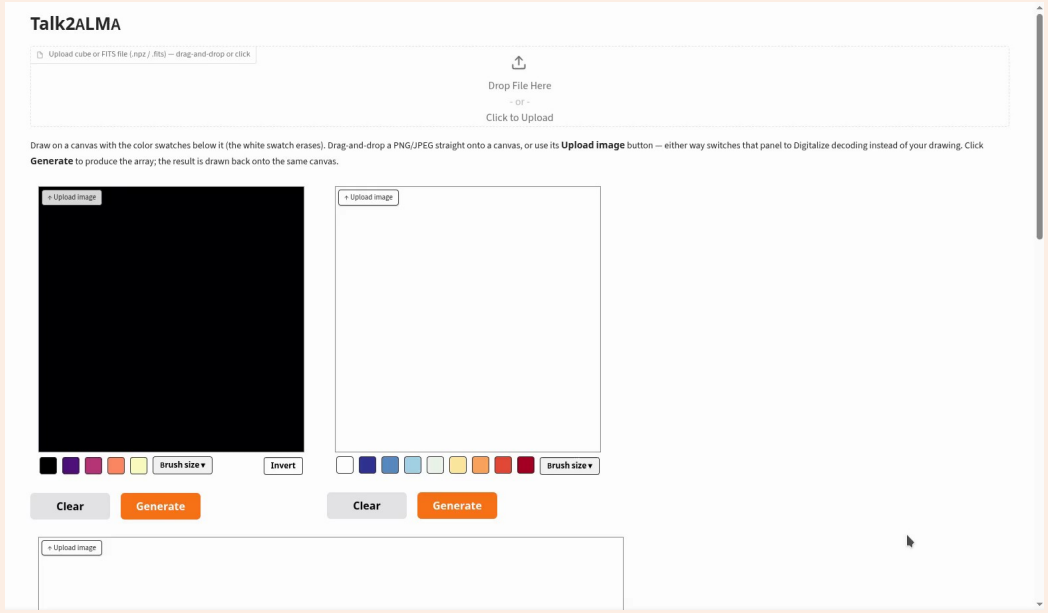
1 Drop a FITS cube (3D)

upload a cube you already have and search with it



2 Draw a velocity field

sketch the pattern by hand and search with it

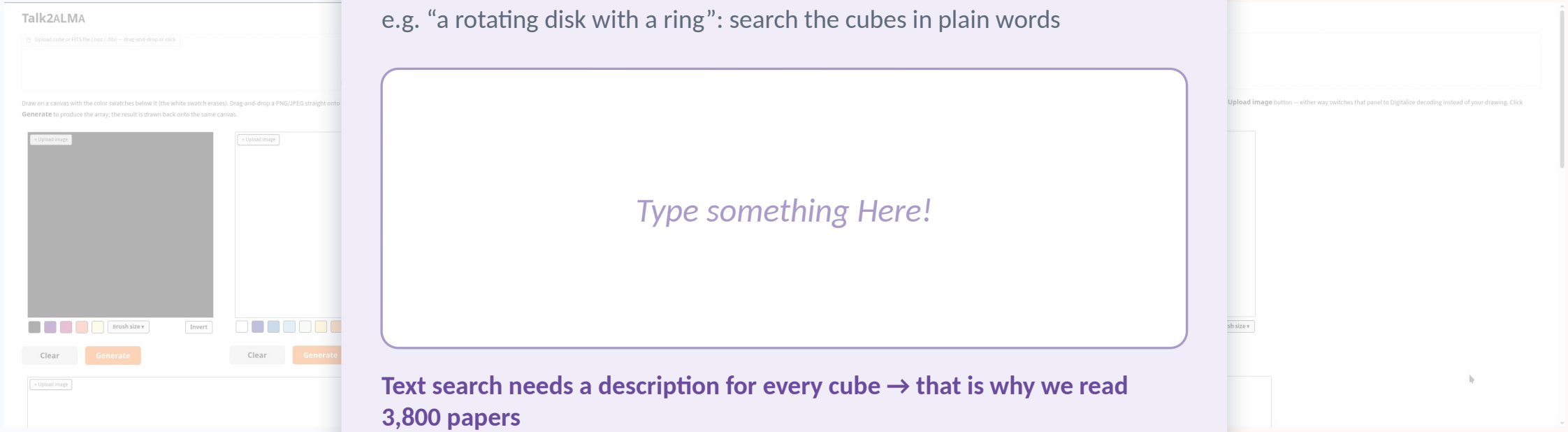


Demo: drop a FITS cube, or draw a velocity field etc

1 Drop a FITS cube (3D)

upload a cube you already have

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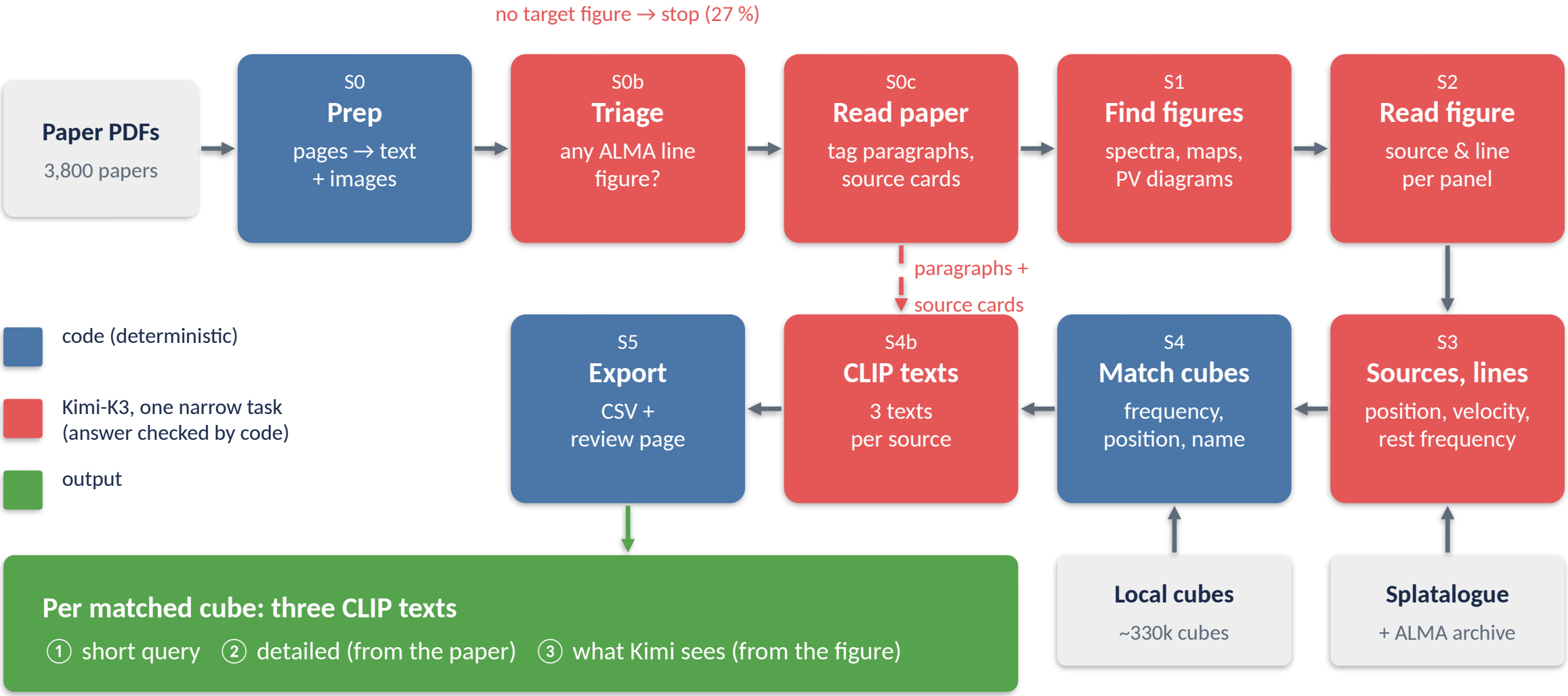
3 Type what you are looking for

e.g. “a rotating disk with a ring”: search the cubes in plain words

Type something Here!

Text search needs a description for every cube → that is why we read 3,800 papers

How the paper agent works



every stage saves its result per paper, so a run can stop and continue · 16 Kimi requests at once

~5 days of API time instead of 1.4 person-years

≥ 2,750 h

by hand

2,750 papers × at least 1 h to find the figures, sources and lines and match the cubes

~5 days

with Kimi-K3 on RAP

16 requests in parallel, day and night; the astronomer checks samples and fixes rules

~60k NTD

total API cost

≈ 18 NTD per paper read in full; a skipped paper costs ≈ 0.3 NTD

~90k

CLIP descriptions

The part no one could do by hand. ~30k figure-panel ↔ cube pairs × 3 texts (short query, detailed, what is seen). Even at 2 min each that is ≥ 3,000 h more, and every text follows the same rules and is checked against the paper.

est. from batch 1 of the full run (200 papers: 147 read in full, 1,750 pairs) · 1 person-year = 2,000 h · RAP price 0.64 NTD per 10k tokens (Nvidia)

Spend tokens only where Kimi is needed

1

Narrow agents, not one big agent

v0 let Kimi explore the whole paper: input grew from 3k to 25k tokens per call, 0 matches after 10 calls. Now each step does one job with a tool budget.

2

Triage first

One cheap call per paper (≈ 0.3 NTD) skips the 27 % with no ALMA line figures. In v5, one NOEMA-only paper used 1/4 of the tokens.

3

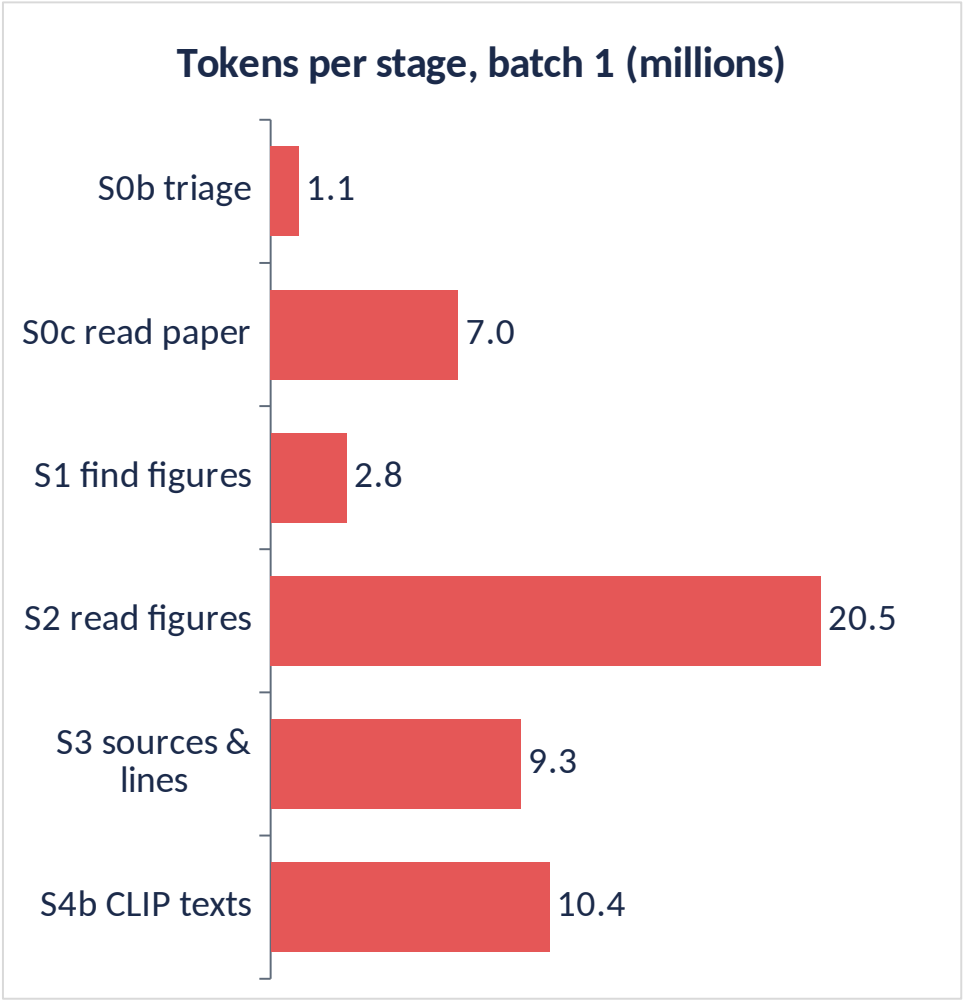
Check the data before calling Kimi

If Splatalogue is down, S3 stops before any call (in v6, 26 papers' tokens were wasted).

4

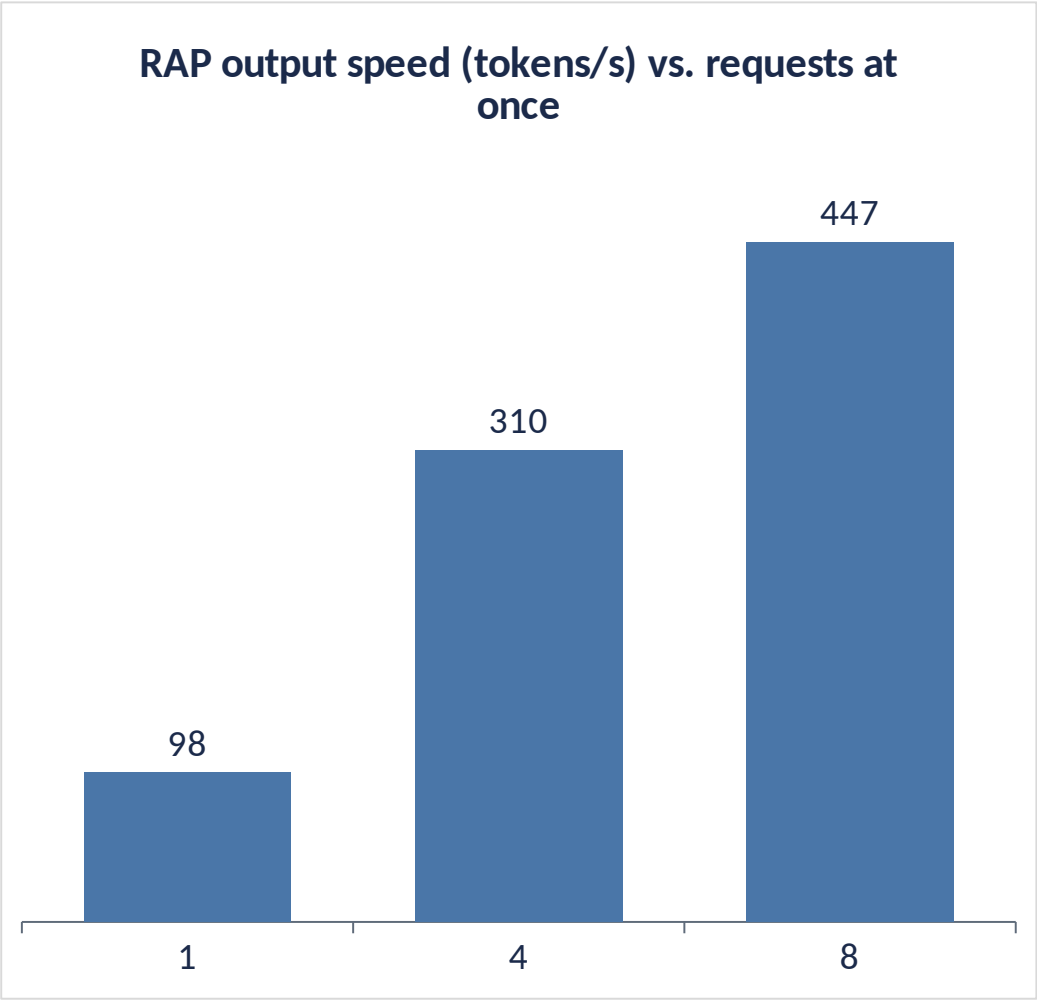
Read each paper once

S0c tags the useful paragraphs once; S2 and S4b reuse them instead of searching again.



batch 1 of the full run: 200 papers, 147 read in full, 51M tokens (36M in, 15M out)

From about 3 weeks to about 5 days



1

Papers in parallel

8 requests at once give 4.5× the output of one request. We run 16 (another +23 %).

2

Pages and figures in parallel

Within a paper too: one paper with 82 figures no longer holds up its whole batch.

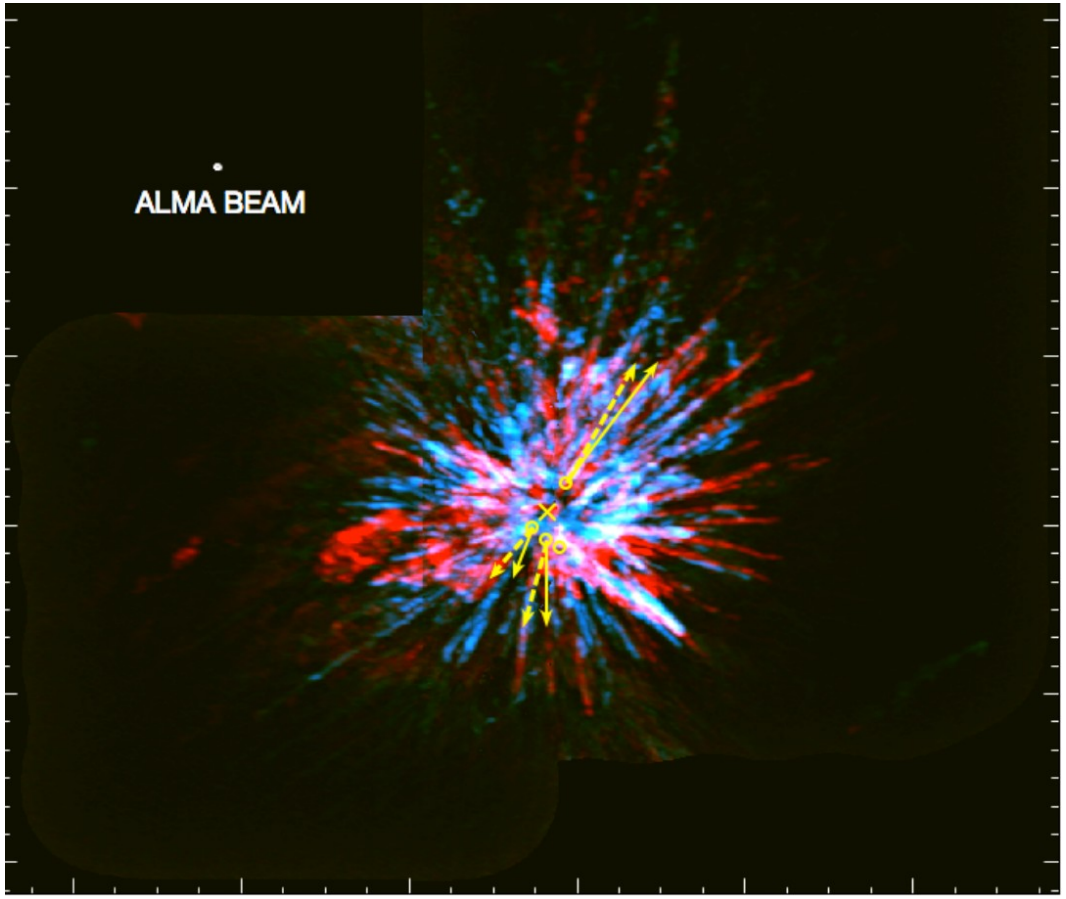
3

Fix the real bottleneck: Splatalogue

S3 waited 161 min per batch with only 1.2 papers active; 43-GHz-wide queries timed out. Now queried in 1-GHz pieces, in the background, before S3 needs them.

one paper at a time: 10–16 h per 100 papers (v6 logs) → 16–25 days for 3,800 · parallel test with unique prompts (test_parallel.py) · full-run time est. from batch 1

Example: the explosive outflow in Orion (OMC1)



Bally et al. 2017, ApJ 837, 60, Fig. 4 (cropped; CO 2-1, blue/red = blue/redshifted)

Matched cube (code, S4)

2013.1.00546.S · OMC1_SE · spw 16

CO 2–1 from Splatalogue; explosion centre inside the mosaic

① short (query-like)

CO 2-1 peak brightness map of OMC1 showing straight clumpy high-velocity streamers from the 500-year-old explosion

② detailed (excerpt)

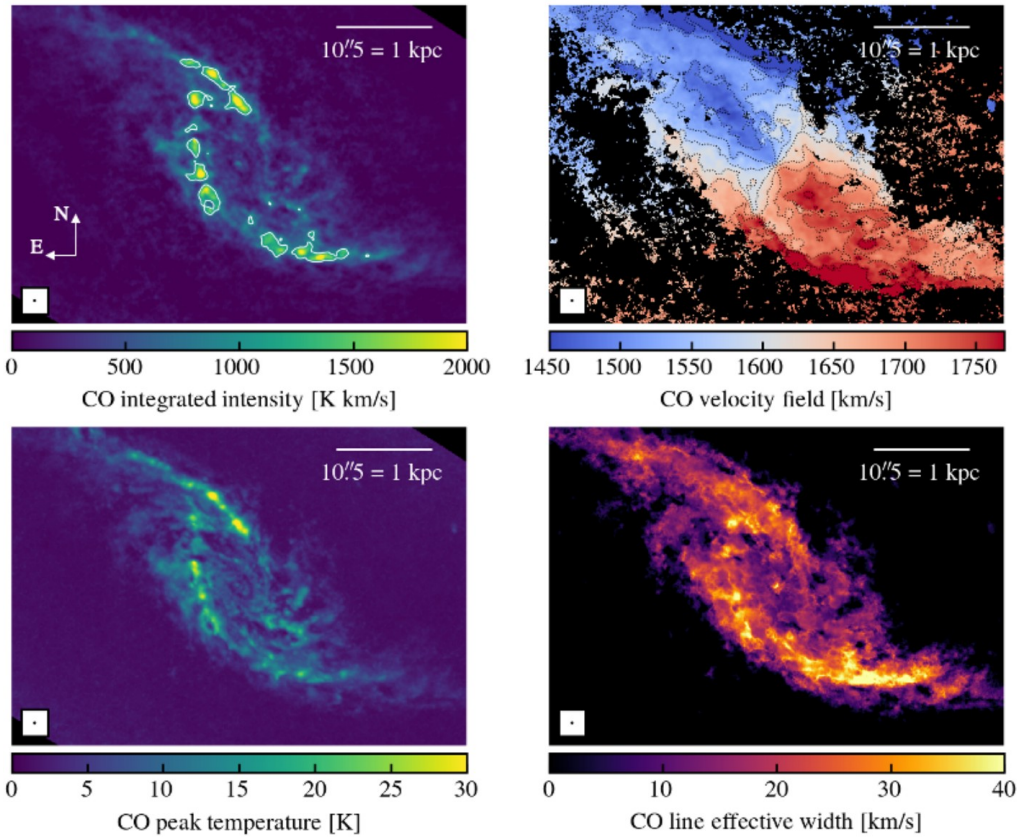
… more than 100 high-velocity streamers that point back to a location between the BN object and Source I. The streamers are straight, barely resolved clumpy filaments … The explosion center measured from the streamer orientations coincides with that from stellar proper motions …

③ what Kimi sees

Over a hundred thin, straight 12CO 2-1 streamers of the OMC1 explosion radiate nearly isotropically from a common center between the BN object and Source I, blueshifted and redshifted streamers intermixed at all position angles. The brightest streamers cluster slightly north-northwest of the ejection center …

from the full run (v8, batch 1) · 26 matched rows for this paper · ③ ends with one more sentence about features near the edges

Example: barred spiral galaxy NGC 1365



Schinnerer et al. 2023, ApJL 944, L15, Fig. 2 (CO 2-1: intensity, velocity, peak, width)

Matched cube (code, S4)

2013.1.01161.S · NGC1365 · spw 16

CO 2–1 from Splatalogue, shifted by the galaxy velocity

① short (query-like)

CO(2-1) integrated intensity map of NGC 1365 revealing bright clumpy molecular bar lanes around a faint smooth inner gas disk

② detailed (excerpt)

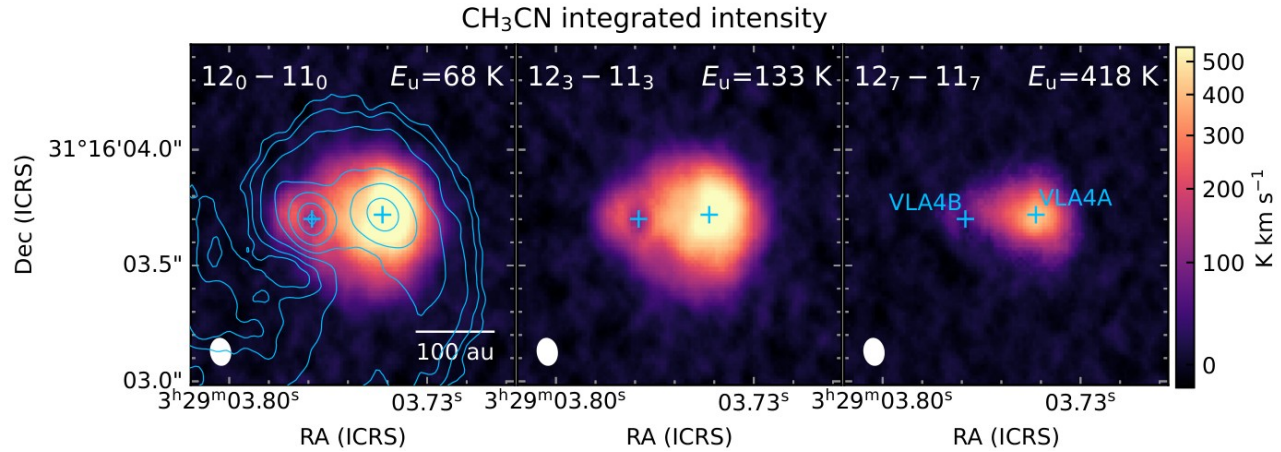
… The integrated intensity over the inner 4.8 × 3.3 kpc² shows bright spiral-like molecular bar lanes surrounding a faint, smooth, lower-brightness gas disk … The lanes are interpreted as bar shocks that funnel gas toward the central disk.

③ what Kimi sees (excerpt)

Extended, clumpy emission organized along two prominent curved lanes (one running roughly north-south through the center, one to the south-west) with about a dozen compact bright peaks slightly larger than the beam; faint smooth diffuse emission fills the interior.

panel 1 shown; the velocity and peak panels match the same cube · ③ shortened (v7 text, before the no-units rule)

Example: protobinary SVS13A



Hsieh et al. 2025, A&A 696, A1, Fig. 2 (CH3CN J=12-11, K = 0, 3, 7)

Matched cube (code, S4)

2022.1.00479.S · SVS13A · spw 33

CH3CN 12(0)–11(0), 220.747 GHz from Splatalogue

VLA4B 0.26″ from the field centre; frequency inside the cube

① short (query-like)

CH3CN 12(0)-11(0) integrated intensity peaking on bursting secondary VLA4A with a ring around primary VLA4B in the SVS13A protobinary

② detailed (excerpt)

… The emission clearly peaks toward VLA4A, while a ring-like structure surrounds VLA4B. The authors interpret the ring as caused by high dust optical depth …

③ what Kimi sees

Around VLA4B, the primary protostar, the emission forms a ring-like structure with a local minimum at the source position, attributed to high dust optical depth; fainter extended emission also appears to the west of the binary.

one of 39 matched rows for this paper; ① ② come from the paper text (S4b), ③ from Kimi looking at the figure (S2)

My workflow: 9 versions in 10 days

1 **Run**

I start the run; Kimi-K3 works through the papers on RAP

2 **Check**

Claude Code and I review the results: figures, matches, texts

3 **Measure**

time and tokens per stage, retries, errors (usage report, logs)

4 **Fix**

Claude Code changes prompts, checks and code; the version is frozen

↻ then the next version

v0	One agent reads the whole paper: too slow, 0 matches
v1	Figure-first, narrow steps, tool budget: 24 matches
v2	Frequencies from Splatalogue, not memory
v3	First CLIP texts, one per panel × molecule
v4	Texts about what each panel shows
v5	10 random papers: caption keyword fixes
v6	Triage step, 100 papers in batches
v7	Read each paper once, 3 texts per cube, 8 requests at once
v8	**Full run: 3,800 papers, 16 requests at once**

2026-10-21 → 2026-10-30 · every version keeps its frozen code and a README with what changed and why

Summary

Talk2ALMA	**Search ALMA data by what they show** query by text, a FITS cube or a sketch; CLIP links cubes to text from 3,800 papers
4.0×	**H100 vs RTX 3080, per GPU** at the same cube size, with 1.95× larger cubes and 2× the batch per GPU
~5 days	**instead of ≥ 2,750 h by hand** Kimi-K3 on RAP reads 3,800 ALMA papers for about 60k NTD (est.)
Beyond speed	**Claude Code as a partner, not only a speed-up** some knowledge came from it, many methods were worked out together: better quality, not just faster

paper-agent numbers est. from batch 1 of the full run (v8); final numbers after all 19 batches